"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh disciplines issues."

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1) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$

$\begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$ $\begin{matrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \end{matrix}$ $\begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{matrix}$

$\det() = 1$ $\det() = 1$ $\det() = 1$

3) a,b) $H^2 = (I - 2uu^T)^2 = I^2 - 4uu^T + 4uu^T u^T$

$u^T u = 1 \quad H^2 = I$

$H = H^T$ and $H^T H = I$ H is an orthogonal (unitary) matrix

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c) $Hu = (I - 2uu^T)u = u - 2uu^T u = u - 2u = -u, \quad \lambda = -1$

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d) $Hv = (I - 2uu^T)v = v - 2uu^T v = v \Rightarrow \lambda = 1$

u has dimension $\rightarrow n-1$

basis of $(n-1)$ orthonormal eigenvectors with this eigenvalue.

Adding the eigenvector u, we find that H is diagonalizable.